《数据库系统原理》实验报告（1）					
题目：练习数据库模式定义语言 DDL					
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实验环境：MariaDB					

实验步骤及结果截图：

1.创建 TeachingManagment 教学管理系统数据库：

```
MariaDB [(none)]> create database TeachingManagment;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| TeachingManagment  |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.002 sec)
```

2.创建 Dept 学院表：

```
MariaDB [(none)]> use TeachingManagment;
Database changed
MariaDB [TeachingManagment]> create table Dept(
    -> No int primary key not null,
    -> Name varchar(50) not null
    -> );
Query OK, 0 rows affected (0.035 sec)

MariaDB [TeachingManagment]> show tables;
+----------------------------+
| Tables_in_TeachingManagment |
+----------------------------+
| Dept                       |
+----------------------------+
1 row in set (0.003 sec)
```

3.创建 Student 学生表：

```
MariaDB [TeachingManagment]> create table Student(
    -> No int primary key not null,
    -> Name varchar(50) not null,
    -> Gender varchar(1) not null,
    -> Age int not null,
    -> D_no int not null,
    -> constraint GHK_Gender check (Gender in('F','M')),
    -> constraint FK_Dept foreign key (D_no) references Dept(No)
    -> );
Query OK, 0 rows affected (0.017 sec)

MariaDB [TeachingManagment]> show tables;
+----------------------------+
| Tables_in_TeachingManagment |
+----------------------------+
| Dept                       |
| Student                    |
+----------------------------+
2 rows in set (0.002 sec)
```

4.创建 Course 课程表：

```
MariaDB [TeachingManagment]> create table Course(
    -> No int primary key not null,
    -> Name varchar(50) not null,
    -> Credit int not null,
    -> D_no int not null,
    -> constraint FK_Course_Dept foreign key (D_no) references Dept(No)
    -> );
Query OK, 0 rows affected (0.038 sec)

MariaDB [TeachingManagment]> show tables;
+----------------------------+
| Tables_in_TeachingManagment |
+----------------------------+
| Course                     |
| Dept                       |
| Student                    |
+----------------------------+
3 rows in set (0.002 sec)
```

5.创建 Score 分数表：

```
MariaDB [TeachingManagment]> create table Score(
    -> S_no int not null,
    -> C_no int not null,
    -> Score int not null,
    -> constraint FK_Score_Student foreign key (S_no) references Student(No),
    -> constraint FK_Score_Course foreign key (C_no) references Course(No)
    -> );
Query OK, 0 rows affected (0.017 sec)

MariaDB [TeachingManagment]> show tables;
+----------------------------+
| Tables_in_TeachingManagment |
+----------------------------+
| Course                     |
| Dept                       |
| Score                      |
| Student                    |
+----------------------------+
4 rows in set (0.002 sec)
```

6.DESC 查询表结构：

```
MariaDB [TeachingManagment]> desc Score;
+-------+---------+------+-----+---------+-------+
| Field | Type    | Null | Key | Default | Extra |
+-------+---------+------+-----+---------+-------+
| S_no  | int(11) | NO   | MUL | NULL    |       |
| C_no  | int(11) | NO   | MUL | NULL    |       |
| Score | int(11) | NO   |     | NULL    |       |
+-------+---------+------+-----+---------+-------+
3 rows in set (0.010 sec)
```

```
MariaDB [TeachingManagment]> desc Student;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| No     | int(11)     | NO   | PRI | NULL    |       |
| Name   | varchar(50) | NO   |     | NULL    |       |
| Gender | varchar(1)  | NO   |     | NULL    |       |
| Age    | int(11)     | NO   |     | NULL    |       |
| D_no   | int(11)     | NO   | MUL | NULL    |       |
+--------+-------------+------+-----+---------+-------+
5 rows in set (0.003 sec)

MariaDB [TeachingManagment]> desc Course;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| No     | int(11)     | NO   | PRI | NULL    |       |
| Name   | varchar(50) | NO   |     | NULL    |       |
| Credit | int(11)     | NO   |     | NULL    |       |
| D_no   | int(11)     | NO   | MUL | NULL    |       |
+--------+-------------+------+-----+---------+-------+
4 rows in set (0.002 sec)
```

```
MariaDB [TeachingManagment]> desc Dept;
+-------+-------------+------+-----+---------+-------+
| Field | Type        | Null | Key | Default | Extra |
+-------+-------------+------+-----+---------+-------+
| No    | int(11)     | NO   | PRI | NULL    |       |
| Name  | varchar(50) | NO   |     | NULL    |       |
+-------+-------------+------+-----+---------+-------+
2 rows in set (0.002 sec)
```

7.查询建立的约束的类型及名字

```
MariaDB [TeachingManagment]> SELECT table_name, constraint_name, constraint_type
    -> FROM information_schema.TABLE_CONSTRAINTS
    -> WHERE table_name = 'Student';
+------------+-----------------+-----------------+
| table_name | constraint_name | constraint_type |
+------------+-----------------+-----------------+
| Student    | PRIMARY         | PRIMARY KEY     |
| Student    | GHK_Gender      | CHECK           |
| Student    | FK_Dept         | FOREIGN KEY     |
+------------+-----------------+-----------------+
3 rows in set (0.009 sec)
```

```
MariaDB [TeachingManagment]> SELECT table_name, constraint_name, constraint_type
    -> FROM information_schema.TABLE_CONSTRAINTS
    -> WHERE table_name = 'Dept';
+------------+-----------------+-----------------+
| table_name | constraint_name | constraint_type |
+------------+-----------------+-----------------+
| Dept       | PRIMARY         | PRIMARY KEY     |
+------------+-----------------+-----------------+
1 row in set (0.019 sec)
```

```
MariaDB [TeachingManagment]> SELECT table_name, constraint_name, constraint_type
    -> FROM information_schema.TABLE_CONSTRAINTS
    -> WHERE table_name = 'Course';
+------------+-----------------+-----------------+
| table_name | constraint_name | constraint_type |
+------------+-----------------+-----------------+
| Course     | PRIMARY         | PRIMARY KEY     |
| Course     | FK_Course_Dept  | FOREIGN KEY     |
+------------+-----------------+-----------------+
2 rows in set (0.008 sec)
```

```
MariaDB [TeachingManagment]> SELECT table_name, constraint_name, constraint_type
    -> FROM information_schema.TABLE_CONSTRAINTS
    -> WHERE table_name = 'Score';
+------------+------------------+-----------------+
| table_name | constraint_name  | constraint_type |
+------------+------------------+-----------------+
| Score      | FK_Score_Course  | FOREIGN KEY     |
| Score      | FK_Score_Student | FOREIGN KEY     |
+------------+------------------+-----------------+
2 rows in set (0.009 sec)
```

8.将 Student 中的 name 字段由 varchar 改为 char（15）

```
MariaDB [TeachingManagment]> alter table Student modify Name char(15) not null;
Query OK, 0 rows affected (0.031 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> desc Student;
+--------+------------+------+-----+---------+-------+
| Field  | Type       | Null | Key | Default | Extra |
+--------+------------+------+-----+---------+-------+
| No     | int(11)    | NO   | PRI | NULL    |       |
| Name   | char(15)   | NO   |     | NULL    |       |
| Gender | varchar(1) | NO   |     | NULL    |       |
| Age    | int(11)    | NO   |     | NULL    |       |
| D_no   | int(11)    | NO   | MUL | NULL    |       |
+--------+------------+------+-----+---------+-------+
5 rows in set (0.004 sec)
```

9.添加 Student 和 Course 的 D_No 列及其约束:

在建表的时候优先建立的是 Dept 表，因此在建立 Student 和 Course 的时候就已经添加了 D_No 列及其约束，如下：

```
MariaDB [TeachingManagment]> create table Student(
    -> No int primary key not null,
    -> Name varchar(50) not null,
    -> Gender varchar(1) not null,
    -> Age int not null,
    -> D_no int not null,
    -> constraint GHK_Gender check (Gender in('F','M')),
    -> constraint FK_Dept foreign key (D_no) references Dept(No)
    -> );
Query OK, 0 rows affected (0.017 sec)

MariaDB [TeachingManagment]> show tables;
+----------------------------+
| Tables_in_TeachingManagment |
+----------------------------+
| Dept                       |
| Student                    |
+----------------------------+
2 rows in set (0.002 sec)
```

```
MariaDB [TeachingManagment]> create table Course(
    -> No int primary key not null,
    -> Name varchar(50) not null,
    -> Credit int not null,
    -> D_no int not null,
    -> constraint FK_Course_Dept foreign key (D_no) references Dept(No)
    -> );
Query OK, 0 rows affected (0.038 sec)

MariaDB [TeachingManagment]> show tables;
+----------------------------+
| Tables_in_TeachingManagment |
+----------------------------+
| Course                     |
| Dept                       |
| Student                    |
+----------------------------+
3 rows in set (0.002 sec)
```

10.将 Score 中的约束删除后再添加一次：

先执行删除

```
MariaDB [TeachingManagment]> SELECT table_name, constraint_name, constraint_type
    -> FROM information_schema.TABLE_CONSTRAINTS
    -> WHERE table_name = 'Score';
+------------+-----------------+-----------------+
| table_name | constraint_name  | constraint_type |
+------------+-----------------+-----------------+
| Score      | FK_Score_Course  | FOREIGN KEY     |
| Score      | FK_Score_Student | FOREIGN KEY     |
+------------+-----------------+-----------------+
2 rows in set (0.008 sec)

MariaDB [TeachingManagment]> alter table Score drop foreign key FK_Score_Course;
Query OK, 0 rows affected (0.025 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> alter table Score drop foreign key FK_Score_Student;
Query OK, 0 rows affected (0.009 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> SELECT table_name, constraint_name, constraint_type  FROM information_schema.TABLE_CO
NSTRAINTS  WHERE table_name = 'Score';
Empty set (0.004 sec)
```

直接重新添加约束发现报错：

```
MariaDB [TeachingManagment]> alter table Score
    -> add constraint
    -> FK_Score_Course foreign key (S_No) references Student(No);
ERROR 1061 (42000): Duplicate key name 'FK_Score_Course'
```

查询资料得知删除约束后数据库中可能还存在索引：

```
MariaDB [TeachingManagment]> show index from Score;
+-------+------------+------------------+--------------+-------------+------
------------+---------+
| Table | Non_unique | Key_name         | Seq_in_index | Column_name | Colla
dex_comment | Ignored |
+-------+------------+------------------+--------------+-------------+------
------------+---------+
| Score |          1 | FK_Score_Student |            1 | S_no        | A
           | NO      |
| Score |          1 | FK_Score_Course  |            1 | C_no        | A
           | NO      |
+-------+------------+------------------+--------------+-------------+------
------------+---------+
2 rows in set (0.010 sec)

MariaDB [TeachingManagment]> ALTER TABLE Score DROP INDEX FK_Score_Course;
Query OK, 0 rows affected (0.013 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> ALTER TABLE Score DROP INDEX FK_Score_Student;
Query OK, 0 rows affected (0.012 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

删除索引后重新添加：

```
MariaDB [TeachingManagment]> ALTER TABLE Score ADD CONSTRAINT FK_Score_Student FOREIGN KEY (S_no) REFERENCES Student(No);
Query OK, 0 rows affected (0.026 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> ALTER TABLE Score ADD CONSTRAINT FK_Score_Course FOREIGN KEY (C_no) REFERENCES Course(No);
Query OK, 0 rows affected (0.008 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [TeachingManagment]> SELECT table_name, constraint_name, constraint_type  FROM information_schema.TABLE_CONSTRAINT
e';
+------------+-----------------+----------------+
| table_name | constraint_name  | constraint_type |
+------------+-----------------+----------------+
| Score      | FK_Score_Course  | FOREIGN KEY     |
| Score      | FK_Score_Student | FOREIGN KEY     |
+------------+-----------------+----------------+
2 rows in set (0.003 sec)
```

遇到的问题：

在删除 Score 中的约束之后直接重新添加约束发现数据库报错

解决方案：

查询资料后得知，在 MariaDB 中，外键约束和索引在命名空间上是共享的，删除了约束系统内部还残留着同名索引，因此导致报错。

因此在删除索引之后重新进行添加，实验成功！